

Hugging Face fuels Johns Hopkins APL's LLM training stack

Johns Hopkins Applied Physics Laboratory (APL) has built a mission-specific large language model training stack powered by open source datasets and tooling from the Hugging Face ecosystem, creating a reusable infrastructure that can help government agencies develop secure, domain-adapted AI systems.

At the centre of the breakthrough is a two-billion-parameter model trained on a specially curated mix of open source datasets spanning general knowledge, mathematics, engineering, and foreign languages. To support this, the APL team developed lightweight in-house infrastructure that blends and streams multiple Hugging Face datasets into manageable chunks while preserving full control, replicability, and transparency across the training workflow.

The new stack was built after existing open source solutions failed to meet all mission requirements, particularly around sensitive data handling, specialised workflows, and deployment in restricted environments. The infrastructure is now available for other researchers to use.

Samuel Barham, artificial intelligence researcher at APL's Intelligent Systems Center, said the capability gives government partners a trusted pathway to build and adapt LLMs for mission-relevant domains while avoiding major technical pitfalls.

The platform is already being used for CBRN threat-response LLMs, global health mission systems, and multimodal warfighting models, with the next phase focused on larger models, richer data types, and broader operational deployment across national security missions.



Microsoft open sources Harrier

Microsoft's Bing team has open sourced Harrier, a family of embedding models released under the MIT license with no usage restrictions, marking the first open source embedding stack to top the multilingual MTEB v2 benchmark. Available on Hugging Face for commercial use, modification, and redistribution, the family supports more than 100 languages and directly outperforms proprietary alternatives from OpenAI, Google, Amazon, and previous NVIDIA leaders.



The release strengthens the view that proprietary embedding models no longer retain a clear quality edge, accelerating the commoditisation of foundation AI infrastructure for enterprise retrieval, RAG, legal discovery, and scientific search.

Harrier ships in three tiers: the flagship 27B model with 25.6B active parameters and 5,376-dimensional embeddings, a 0.6B mid-range version, and a 270M edge model. All three share the same API, embedding format, and a 32,768-token context window, enabling a consistent path from prototyping to production-scale deployment.

Microsoft also plans to integrate Harrier across Bing, grounding services, and AI agent memory, ranking, and orchestration layers. The Bing team described grounding as "the foundational capability that drives user trust for any AI agent," positioning Harrier as "a foundational layer for memory, ranking, and orchestration" in the emerging agentic web.

FossID's Agentic SCA brings checks into AI coding

FossID has unveiled Agentic SCA, a new software composition analysis layer purpose-built for AI-driven software development, marking a major shift in how open source compliance is enforced within modern coding workflows.



Designed for an era where AI tools increasingly generate and modify code from fragmented sources, the new capability embeds real-time identification of open source, third-party, and proprietary code directly into the development process, including whole files and snippet-level generated or copied code.

The system automatically flags licence obligations, mixed-licence conflicts, copyright exposure, known vulnerabilities, and remediation pathways, enabling developers and AI agents to make compliant decisions before code is committed.

By moving compliance checks upstream, FossID shifts OSS governance from delayed post-build reviews to continuous policy enforcement at code creation itself, significantly improving SBOM accuracy and continuous software supply chain visibility as code evolves.

Agentic SCA also transforms software audits through signature scanning, snippet detection, dependency analysis, deep licence and copyright review, and structured shareable reports, while its MCP Server, Skills, and Hooks architecture makes FossID's audit intelligence directly accessible to AI agents.

The platform is already in pilot with enterprise users across automotive, semiconductor, telecom, and software, with broader availability expected in H2 2026.